A069 (provisional) — 3D Swirl Clock Emergence

v0.2.0: exact Euler controls and framework v1.0.9

SST-Workbench — source-qualified methodological programme

17 October 2026 — development control campaign

Abstract

The physical target is an independently predicted internal mode frequency from qualified three-dimensional vortex dynamics, without a prescribed time dilation factor. The v0.2.0 execution covers only two *analytic numerical controls*: D1, an exact unbounded affine three-dimensional Euler flow inherited from v0.1.0; and D2A, a new steady smooth columnar vortex (two-dimensional motion embedded in three-dimensional space). Neither tests a finite-length knotted Euler vortex or an atomic Swirl Clock. The test instance is generated with framework v1.0.9 and conservatively prevents physical PKLSA and relativity claims. The A069 registry slot remains provisional.

1 Research question and scientific scope

Can an independently sourced `DYNAMICS_READY` finite-core 3D PKLSA vortex support a gauge-invariant recurrent internal mode whose angular frequency is predicted from initial hydrodynamic fields, not from a predetermined Lorentz factor or observed clock frequency? A negative result on preregistered physical holdouts would reject the proposed frequency closure *within that qualified dynamical model*. D1 and D2A provide implementation qualification only; they cannot settle this question.

Competing hypotheses. H_0: no independently identifiable, numerically convergent recurrent physical phase exists, or the frozen predicted rate fails held-out provider comparisons. H_1: a stable phase is identified without fitting the target, and the frozen rate matches independent holdouts with signed, uncertainty-qualified agreement. Both physical hypotheses remain `UNRESOLVED`. The analytic reference cases are not statistical samples of distinct physical vortex systems.

2 Assumptions and boundary conditions

The source medium is considered inviscid, barotropic and incompressible only within each specified exact control. D1 has infinite energy in $\mathbb{R}^3$. D2A has a smooth finite core and finite kinetic energy *per unit axial length*, but is invariant along z, so its full three-dimensional energy is infinite. Both controls have known trajectories by construction. No knot eigenmode or physical time flow is inferred from a Lagrangian tracer's azimuth. E010-v0.3.1/E011-v0.3.0 are admitted only as upstream geometry candidates, not as dynamics or clock data. Legacy KnotPlot visually relaxed exports and VortexLab proxy runs are not admitted as physical science sources.

3 Symbols, dimensions and exact reference equations

Symbol	Physical meaning	SI dimension
T	absolute simulation time	s
$\mathbf{u}$	continuum velocity	m/s
R, a	material loop radius; column width	m
Ω, C, ω	angular rates / vorticity	s^{-1}
Γ	circulation	m^2/s
ρ, p	density; pressure	$\mathrm{kg/m}^3$; Pa
θ	measured tracer angle (not proper time)	rad (dimensionless)

3.1 D1: affine vortex stretching and Kelvin circulation

$$\mathbf{u} = A(T)\mathbf{x}, \quad A = \begin{pmatrix} -s/2 & -\Omega & 0 \\ \Omega & -s/2 & 0 \\ 0 & 0 & s \end{pmatrix}, \quad \Omega(T) = \Omega_0 e^{sT}, \quad R(T) = R_0 e^{-sT/2}. \tag{1}$$

Its exact 3D incompressible Euler pressure Hessian is $\nabla\nabla p = -\rho\,\mathrm{sym}(\dot{A} + A^2)$. The Kelvin invariant is $\Gamma = 2\pi R^2\Omega = \text{const}$; the independently integrated Cartesian trajectory is compared with $\theta(T) = \Omega_0(e^{sT} - 1)/s$, using the continuous $s = 0$ limit.

3.2 D2A: smooth spatially inhomogeneous columnar Euler vortex

$$\mathbf{u} = (-y\Omega(r),\ x\Omega(r), 0), \quad \Omega(r) = Ce^{-r^2/a^2}, \quad r = \sqrt{x^2 + y^2}, \tag{2}$$

$$\omega_z = 2C\left(1 - \frac{r^2}{a^2}\right)e^{-r^2/a^2}, \quad \Gamma(r) = 2\pi r^2\Omega(r), \tag{3}$$

$$\frac{dp}{dr} = \rho r\Omega(r)^2, \quad p(r) - p(0) = \frac{\rho C^2 a^2}{4}\left(1 - e^{-2r^2/a^2}\right), \tag{4}$$

$$\frac{E_{\mathrm{kin}}}{L_z} = \frac{\pi\rho C^2 a^4}{8}, \qquad \theta_{\mathrm{pred}}(T; r_0) = Ce^{-(r_0/a)^2}T. \tag{5}$$

The final line gives the phase of *one fluid tracer* at fixed r_0, not the proper time of an atom. The Cartesian RK4 and midpoint solvers never receive θ_{pred} or a relativistic slowdown factor. They integrate $\dot{\mathbf{x}} = \mathbf{u}(\mathbf{x})$ and reconstruct angle with an unwrapped $\mathrm{atan2}(y, x)$. The finite-difference Euler, curl and pressure gradients are independently evaluated. A superposed rigid background rotation is a reference-frame decomposition check, not an extra localized particle source.

4 Algorithm, reproducibility and source qualification

Step 1: verify the preregistered byte-exact D1 inherited source hash and exclude legacy visualization inputs. Step 2: regenerate the D1 exact Euler/Kelvin/control phase and compare with the stored v0.1.0 reference. Step 3: evaluate D2A analytically at both orientations and three independent radii. Step 4: integrate Cartesian trajectories using RK4 at 160, 320 and 640 steps and midpoint at 2560 steps with fixed final time. Step 5: independently finite-difference velocity and pressure and compare the residuals with fixed thresholds. Step 6: report D2B and D3 as unrun, without training on or manufacturing their data.

Provenance. The A069-v0.1.0 D1 source SHA-256 is checked against `configs/BLIND_ANALYTIC_CONTROL_V020.json`. It remains the same analytic-control source, not a new independent scientific provider. E010/E011 source metadata and contracts are included as constraints, not as an assertion that new E011 physical trajectories were tested.

5 Gate structure and interpretations

Gate	Test	Science interpretation
G0–G3	Framework, exclusions, D1 Euler and material tracer	Analytic control only
G4–G9	Full 3D dynamics, internal mode, qualified PKLSA and holdout	`UNRESOLVED`/disabled
G10–G11	Native physical dynamics and A061 handoff	`UNRESOLVED`
G12	Smooth columnar stationary Euler field and pressure	Analytic control only
G13	Tracer time-step convergence and second integrator	Analytic control only
G14	Exact parent D1 hash and fresh analytic regeneration	Analytic control only

No physical gate can `PASS` by proxy because an analytic gate passed. Framework minimal profile may report its disabled G4–G7 as `NOT_APPLICABLE`, which never denotes scientific closure.

Numerical thresholds and limits. The D2A control uses a fixed final time, two signed rates, three normalized radii, a central Cartesian derivative step $10^{-5}a$, an exact pressure law, and distinct RK4/midpoint integrators. Acceptance: phase error below 2×10^{-7} rad; normalized radial error below 2×10^{-7}; momentum, divergence and curl absolute residuals below 2×10^{-8}; pressure relative error below 3×10^{-9}. These are analytic-control precision tolerances and do not represent uncertainties of a physical SST particle. The full three-dimensional spatial resolution, core-model and pressure-Poisson convergence remain untested.

Execution and backend authority. Python NumPy FP64 is `ANALYTIC_CONTROL_ONLY`; no C++/SYCL physics backend is certified by these calculations. Framework v1.0.9 has official 72/72 Linux regression validation in its independent upstream release record, but target Windows/Arc A770 backend parity is pending. Framework *source* v1.0.9 retains scientific kernels/thresholds from v1.0.7; migration changes immutability of run history, report publication and proof bookkeeping.

6 Blindness, replication and scientific hygiene

D1 and D2A are fully published *positive controls* and are not meaningful blind physical experiments. The actual PKLSA mode rate and reveal targets are absent. The official framework still seals and hashes preregistered files and produces BLIND provenance artifacts. A full physics claim requires future DYNAMICS_READY provider files with opaque IDs and preregistered held-out signed frequency predictions. No reveal may be approved while physical gates are unresolved.

Independence. Two signs, multiple radii, RK4 step refinements, and replays are correlated methodological checks, not four independent physical observations. No p-value, Bayes factor, or detection significance is meaningful for D1/D2A. New E011 physical providers must be stratified by actual upstream family, not cloned geometries, common numerical generators or mesh size.

7 Executed results and automatic provenance

The original D1 reference is embedded as read-only input and is rechecked against a SHA-256 before any generated comparison. The new D2A computation is independently run using the declared equations; its results are inserted below by framework v1.0.9 after execution.

Automatic run provenance

Field	Recorded value
Run ID	20261017T220651724812Z-d1413cf7
Run phase	BLIND
Protocol bundle SHA-256 (byte exact)	55dca109c8c31c48b64771afc2660754664adcb0b7943b111855206ba196354c
Portable protocol diagnostic SHA-256	abaff8f8cc00515a774db848944b8024dce9b1fb0035d5e2d655d49c1c5174fd
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Science evidence SHA-256	5587d243bfbaf8f6fbc5cb2f3a54c613e95f56b4c6cdb4da16d9e8b3b63c8466
Final output manifest	See OUTPUT_MANIFEST.json (finalized after PDF and report metadata)

Gate ledger

Gate	Status	Question	Reason
G0	PASS	Is official framework protocol integrity verified?	Official v1.0.9 framework verified frozen byte-exact protocol before science hook.
G1	PASS	Were historical KnotPlot and unqualified visual relaxations excluded from scientific D1 inputs?	D1 inherited synthetic analytic control only; old KnotPlot/VortexLab inputs not admitted.
G2	PASS	Does exact affine 3D Euler control satisfy divergence, Euler, Kelvin and vortex-stretching identities?	D1 exact affine Euler and Kelvin identities.
G3	PASS	Does a Lagrangian Cartesian tracer yield the independently predicted control angular phase?	D1 Cartesian phase identity; not an SST clock.

G12	PASS	Does the smooth finite-core columnar exact Euler control satisfy pressure, curl, incompressibility and phase tests?	Exact smooth stationary 2D columnar Euler embedded in 3D, finite energy per axial length, not knotted dynamics.
G13	PASS	Do two time-integrators and the 160/320/640 refinement ladder reproduce the analytic columnar phase without clock fit?	Independent Cartesian tracer solver, RK4 vs midpoint and step refinements, not a clock eigenmode.
G14	PASS	Does the inherited D1 dataset match the preregistered original SHA-256 and independently regenerated D1 reference?	Inherited analytic dataset byte hash compared with frozen SHA and regenerated D1 reference; no new experimental source.
G4	NOT_APPLICABLE	Is a true finite-core 3D Euler vortex trajectory dynamically qualified?	Minimal framework profile disables G4; No fully 3D, finite-length, finite-core PKLSA Euler trajectory; D2A is columnar only.
G5	NOT_APPLICABLE	Is there a stable internal mode phase independent of mesh gauge and background rotation?	Minimal framework profile disables G5; No independently identified stable knot-internal material phase observable.
G6	NOT_APPLICABLE	Are E010/E011 origins SHA-verified and dynamically qualified rather than only STATIC_READY?	Minimal framework profile disables G6; E010/E011 real SHA-verified dynamical provider not supplied; STATIC_READY is not DYNAMICS_READY.
G7	NOT_APPLICABLE	Does an independently precommitted rate prediction match the held-out mode phase?	Minimal framework profile disables G7; No preregistered physical clock-rate holdout predictions; analytic azimuth is only a tracer.
G8	UNRESOLVED	Do temporal/spatial/core and pressure-Poisson convergence, energy and helicity checks pass?	No full 3D pressure-Poisson/energy/helicity/core and long-time convergence qualification.
G9	UNRESOLVED	Does provider-independent PKLSA holdout confirm signed rate and uncertainty without refitting?	No provider-independent PKLSA physics holdout; physical reveal forbidden.
G10	UNRESOLVED	Has actual strict C++ FP64 certification executed for the physical numerical kernel?	No independently compiled physical C++ FP64 dynamical kernel (minimal reference profile).
G11	UNRESOLVED	Is downstream A061 transfer permitted by all upstream science qualification gates?	A061 science handoff prohibited without G4-G10.

Backend manifest


```
{
  "D1_status": "PASS",
  "D2A_status": "PASS",
  "actual_backend": "python-numpy-fp64",
  "authority": "ANALYTIC_CONTROL_ONLY",
  "framework_target": "1.0.9",
  "no_native_backend_claim": true,
  "old_dataset_sha256":
"29ba1012aa3c20a3d94cf0e9898578aa0385d8f0c0986d490426035c0952f0dc",
  "physical_clock_prediction": false,
  "physical_pklsa_status": "UNRESOLVED",
  "precision": "float64",
  "schema": "SST-BACKEND-MANIFEST-2"
}
```


Source manifest


```
{
  "manifest_sha256":
"4b0290731a8e962bf9481afed200429fa7f5427f32f32b1ff7beedd51824a7da",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A069",
      "evidence_class": "synthetic_control",
      "exists": true,
```

```
      "independence_family": "A069_D1_same_equations",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifi
ers\\A069_3D_SwirlClock_Emergence\\A069-v0.2.0\\sources\\A069_v0.1.0\\D1_ANAL
YTIC_CONTROL.json",
      "provenance_family": "A069_v0.1.0_self_generated",
      "registry_hint": {
        "catalog_id": "A069",
        "hits": []
      },
      "role": "analytic_regression",
      "sha256":
"29ba1012aa3c20a3d94cf0e9898578aa0385d8f0c0986d490426035c0952f0dc",
      "source_id": "S0",
      "version": "v0.1.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench"
}
```


8 Interpretation and reserved physical next steps

Passing D1 or D2A is a *software and exact-solution numerical qualification*, not a demonstration of time dilation, an emergent atomic clock, independent PKLSA knot stability, monometricity, or gravity. D2B must first construct and evolve finite-energy, finite-core genuinely 3D fields with validated pressure projection, nonzero axial structure, material vs mesh phase, converged energy/helicity/circulation, time and mesh refinement, and physically meaningful mode lifetime. D3 requires independently held-out E011 providers and an outcome-blind transfer law. A061 downstream clock data cannot be interpreted as SST until that upstream physics closure exists.

9 Execution commands and framework retention

```
python -m unittest discover -s tests -v
run_all.cmd FREEZE
run_all.cmd BASIC
run_all.cmd REPORT_ONLY
run_all.cmd REVEAL_IF_ALLOWED
```

The instance is pinned by `.sst_framework_root` to the sibling framework v1.0.9 under `06_templates`, with no permitted silent fallback to v1.0.7. Exact source, report and science contracts are frozen before the official run. Nonce-backed reveal material remains private.

References

[1] W. Thomson (Lord Kelvin), "On Vortex Motion", *Trans. Royal Society of Edinburgh* **25** (1869), 217–260.

[2] P. G. Saffman, *Vortex Dynamics*, Cambridge University Press (1992).

[3] SST-Workbench, E010 PKLSA production knot/link basis v0.3.1, https://drive.google.com/drive/folders/14sNZOxYoNkI4a-j2HxPoC51xXb0XwVy7.

[4] SST-Workbench, E011 static ready seed atlas v0.3.0, https://drive.google.com/drive/folders/1lnHSQxcmQwvlde4WEJo46CGkoJQxXnlH.

[5] SST Falsifier Framework v1.0.9, CANONICAL_FROZEN, validation (2026-10-17).

[6] SST, CANON-v0.8.36, §3.14 Chronos–Kelvin invariant, https://drive.google.com/file/d/15n8i5dWzHSGA72fU4LZoryRrj7BF3n9n/view.